

Nicholas Channg

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EDUCATION

Cornell University

Ithaca, NY

B.A. Computer Science, Minors in Artificial Intelligence, Data Science | GPA: 3.7/4.0 Expected Graduation, May 2027

- **Awards:** Y Combinator AI Startup School 2025 (1 of 2,500 selected), 4x Hackathon Winner

EXPERIENCE

Google

Sunnyvale, CA

Software Engineer Intern, Cloud Knowledge Catalog

May 2026 – Present

- Engineered within two backend microservices in the Knowledge Catalog data governance platform to deliver lightweight data profiling on BigQuery external tables and views, surfacing data insights that ground Gemini Enterprise AI agents
- Extended a Java microservice handling end-user API requests to enable low-latency, cost-efficient lightweight profiling for BigQuery external tables and views, rolled out safely behind Mendel feature flags
- Developed dynamic query-generation logic in the Go Orchestrator that selects profiling strategies by table format, applying TABLESAMPLE or a LIMIT fallback to avoid full-table scans, computing sampling rates from BigQuery fetched row-counts

Amazon

New York, NY

Software Engineer Intern, Amazon Ads

Sep 2025 – Dec 2025

- Developed and owned asynchronous Java and Python backend services in a distributed AWS video ads processing pipeline handling 80K+ jobs weekly, designing fault-tolerant workflows across SQS, ECS, Lambda, DynamoDB, and S3
- Built a production AI microservice on AWS Bedrock with a two-stage VLM-to-LLM pipeline to perform multimodal video analysis on video ads, generating titles/descriptions that outperformed advertiser-supplied text in 98% of cases
- Integrated a video assembly service into the pipeline by building a Lambda-triggered ECS execution path and S3/DynamoDB asset storage, enabling secure cross-account and cross-service access of final video assets

Tesla

Fremont, CA

Software Engineer Intern, Charging Team

May 2025 – Sep 2025

- Developed scalable REST API endpoints in the Tesla Mobile App backend with Node.js, Express.js, and TypeScript, orchestrating vehicle/charging microservice calls to power real-time user-visible data for 3.4M daily users
- Implemented a Redis caching layer to store Spark API responses for supercharger congestion/pricing data, reducing redundant upstream microservice calls and improving Tesla Mobile App response latency by 10%
- Developed an agentic AI QA assistant with Python that automated mobile app test-account setup (e.g. region switching, vehicle linking) via orchestrated microservice API calls, improving QA throughput by 25%

Flow

Remote

Software Engineer Intern

Jan 2024 – May 2024

- Developed REST APIs using Flask and Python to retrieve and process data from Flow's AI models, automating customer prospecting and increasing sales representative call volume by 300% (30 → 120 calls/day)
- Reduced application latency by 15% by optimizing PostgreSQL queries; replaced nested SELECTs with JOINs, added B-tree indexes on high-cardinality columns, and batched execution to minimize database calls
- Integrated backend services into CI/CD pipelines using GitHub Actions, ensuring automated testing and deployment

PROJECTS

ElectAI | Python, Scikit-learn, Flask, React, TypeScript, TailwindCSS, NumPy, Pandas

Jan 2025

- Engineered a full-stack ML web app to predict voter turnout using historical behavioral data patterns to model participation likelihood using a Flask REST API to retrieve data from the model and a React frontend
- Built a data preprocessing pipeline with Python, NumPy, and Pandas to clean 20 years of election data, enabling accurate regression modeling with Scikit-learn and achieving high accuracy (RMSE: 5.35, MAE: 4.22)

TECHNICAL SKILLS

Languages: Java, Go, Python, C, C++, TypeScript, JavaScript, SQL

Frameworks: Flask, FastAPI, Node.js, Express.js, React, Next.js

Cloud/Infrastructure: AWS, Redis, PostgreSQL, Docker, Github Actions, CI/CD

Libraries/Tools: Scikit-learn, NumPy, Pandas, OpenAI API, TailwindCSS, Git, Postman